
Regularization & Model Selection

ECON 8310: Business Forecasting — Lecture 6

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Lecture Outline

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- 2 Shrinkage Methods
- 3 Choosing the Penalty
- 4 In Practice
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The Problem: Too Many Predictors

Feature engineering gives us more predictors than we can safely estimate. Something has to decide which ones survive.

Why Regularize?

OLS minimizes $\sum_i (y_i - \mathbf{x}'_i \boldsymbol{\beta})^2$. A weekly model with lags, calendar dummies, promotions, and macro series can easily carry $k = 60$ predictors on $n = 150$ observations — OLS will fit that sample beautifully and forecast badly.

Three things go wrong at once. The variance of $\hat{\boldsymbol{\beta}}$ grows with k . Correlated predictors produce unstable, wildly large coefficients with opposing signs. And once $k > n$ there is no unique solution at all. **Regularization** adds a penalty on coefficient size:

$$\underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{fit}} + \underbrace{\lambda \cdot \text{Penalty}(\boldsymbol{\beta})}_{\text{complexity}}$$

You accept a little bias to buy a large reduction in variance — and for a forecast, that trade is almost always worth making.

Subset Selection: The Classical Answer

Before shrinkage, the standard approach was to *count* predictors rather than shrink them (Guyon and Elisseeff 2003).

Best subset fits all 2^k models and picks the winner — exact, and computationally hopeless past $k \approx 30$. **Stepwise** adds the strongest predictor or drops the weakest, one at a time; it is greedy, and can miss the best combination entirely. Either way the choice is made by adjusted R^2 , AIC, AICc, BIC, or CV.

Subset selection is **discrete** — a predictor is in or out. Small changes in the data flip predictors in and out, so the *selected model* is itself high-variance. You have not removed the instability; you have moved it somewhere harder to see.

This is the approach in *FPP* §7.5. Shrinkage replaces the in-or-out decision with a continuous dial — more stable, and easier to tune.

Shrinkage Methods

Keep every predictor, but pull its coefficient toward zero.

Ridge Regression (ℓ_2)

Ridge (Hoerl and Kennard 1970)

$$\hat{\beta}^{\text{ridge}} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2 + \lambda \sum_{j=1}^k \beta_j^2$$

Ridge shrinks every coefficient toward zero but sets none exactly to zero. At $\lambda = 0$ you have OLS; as $\lambda \rightarrow \infty$ all $\beta_j \rightarrow 0$. Its particular strength is correlated predictors: rather than letting two collinear variables fight for the same coefficient, it splits the effect between them.

Use it when you believe most predictors carry a little signal and you want to keep them all.

Standardize first. The penalty is not scale-invariant — a predictor measured in dollars is penalized differently from the same predictor in thousands of dollars. Get this wrong and the units decide which variables survive.

LASSO (Tibshirani 1996)

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2 + \lambda \sum_{j=1}^k |\beta_j|$$

Swap the squared penalty for an absolute one and the behavior changes qualitatively. LASSO sets some coefficients **exactly** to zero, performing selection and estimation in a single step, and leaving a sparse model you can actually read aloud to a stakeholder.

Use it when you believe only a handful of predictors genuinely matter and you want the model to tell you which.

But be careful what you conclude. Among a group of highly correlated predictors, LASSO keeps *one* more or less arbitrarily and zeros the rest. The survivor is not “the important one” — it is the one that happened to win a coin toss between near-identical variables.

Why LASSO Zeros Out and Ridge Does Not

Both solve “minimize squared error subject to a budget on coefficient size.” The *shape* of the budget is the whole difference.

Ridge: $\sum_j \beta_j^2 \leq t$ — a **circle**

- Smooth boundary, no corners
- Error contours almost never touch it on an axis
- So coefficients get small, never zero

LASSO: $\sum_j |\beta_j| \leq t$ — a **diamond**

- Corners sit exactly on the axes
- Contours are likely to hit a corner first
- A corner means $\beta_j = 0$ exactly

So the sparsity of LASSO is a geometric accident of the ℓ_1 ball having corners — not a statistical test of significance. A zeroed coefficient means “not worth its budget here,” not “proven irrelevant.” See James et al. (2023) §6.2.

Elastic Net: Both Penalties at Once

Elastic Net (Zou and Hastie 2005)

$$\sum_{i=1}^n (y_i - \mathbf{x}_i' \boldsymbol{\beta})^2 + \lambda \left[\alpha \sum_j |\beta_j| + (1 - \alpha) \sum_j \beta_j^2 \right]$$

λ sets the total penalty; $\alpha \in [0, 1]$ sets the mix ($\alpha = 1$ is LASSO, $\alpha = 0$ is Ridge).

The mix keeps LASSO's ability to zero out predictors while adding Ridge's **grouping effect**: correlated predictors now enter or leave together instead of one surviving arbitrarily.

This is usually the right default for forecasting. Lagged predictors are correlated almost by construction — y_{t-1} and y_{t-2} move together — which is precisely the situation where pure LASSO picks a winner at random and pure Ridge refuses to simplify.

Choosing the Penalty

λ is not estimated from the model. It is tuned — and how you tune it decides whether the model generalizes.

The Regularization Path

Fit the model across a grid of λ values and trace each coefficient as it moves.

Read the path left to right. At small λ the coefficients sit near their OLS values and the model is dense and unstable. As λ grows they shrink, and under LASSO they drop to zero one at a time. At large λ only the most robust predictors survive, and in the limit you are left with the intercept — the mean forecast. **The order in which predictors leave is the useful output.** The last survivors are the ones that hold up under the most pressure. Report that ordering rather than a single model's coefficient table — far more stable across resamples, and far more honest about what the data can support.

Choosing λ by Cross-Validation

λ is not estimated from the model — it is tuned, and how you tune it decides whether the model generalizes.

1. Define a grid of λ , usually log-spaced
2. For each λ , run walk-forward CV and record the mean validation RMSE
3. Pick the λ minimizing CV error, then refit on all training data

The one-standard-error rule is worth knowing: rather than the outright minimum, take the *largest* λ whose CV error is within one standard error of it. You give up almost no accuracy and get a simpler, more stable model.

Never use KFold here. Random folds put future observations in the training set and past ones in validation. The CV error will look excellent and the forecast will fail (Bergmeir et al. 2018). Use `TimeSeriesSplit`, exactly as in Lecture 1's walk-forward evaluation.

In Practice

Standardize, tune on a time-aware split, and read the path.

Code Template

```
from sklearn.linear_model import ElasticNetCV
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import TimeSeriesSplit

model = make_pipeline(StandardScaler(),
ElasticNetCV(l1_ratio=[.1,.5,.7,.9,.95,1],
cv=TimeSeriesSplit(5), random_state=42)).fit(X_tr, y_tr)
```

`make_pipeline` is the part that matters: it keeps standardization *inside* each CV fold. Scale on the full sample and you have leaked the validation set into training, and your CV error is optimistic.

`l1_ratio` is α , the Ridge/LASSO mix; the `...CV` variants tune λ for you.

`sklearn` calls the penalty `alpha`, not `lambda`.

Key Takeaways and Roadmap

Shrinkage is the linear-model counterpart to what trees do by pruning.

Linear Methods: Which One When

Method	Penalty	Zeros out?	Best when
OLS	none	no	$k \ll n$, predictors uncorrelated
Stepwise	discrete	yes	Few candidate predictors, need a story
Ridge	ℓ_2	no	Many predictors, all mildly useful
LASSO	ℓ_1	yes	Few predictors truly matter
Elastic Net	both	yes	Many <i>correlated</i> predictors

Connection to Lecture 5. Trees and shrinkage solve the same problem in different geometries. XGBoost's `reg_lambda` is literally a Ridge penalty on leaf weights, and its `gamma` penalizes leaf count the way LASSO penalizes nonzero coefficients. Regularization is not a linear-model topic — it is how every flexible model is kept honest.

Lecture 6: Key Takeaways

1. With many predictors OLS overfits: coefficient variance grows with k , and with $k > n$ there is no unique solution at all.
2. **Subset selection** counts predictors in or out. It is discrete, greedy, and unstable — small data changes flip the selected model.
3. **Shrinkage** replaces that switch with a dial. Ridge (ℓ_2) shrinks everything smoothly; LASSO (ℓ_1) shrinks and zeros out; Elastic Net does both and keeps correlated predictors together.
4. **Standardize first**, inside each CV fold — none of these penalties are scale-invariant. λ **is tuned, not estimated**: use `TimeSeriesSplit`, never random K -fold, which leaks the future.
5. The **regularization path** — the order in which predictors drop out — is a more stable finding than any single coefficient table.

Next: Introduction to Neural Networks (PyTorch) — where these same penalties reappear as weight decay and dropout.

References I

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